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- Average acceleration formula: $\vec{a}_{av} = \frac{\vec{v}_f - \vec{v}_i}{\Delta t}$
- Area of trapezoid in V-t graph

$\vec{\Delta d} = \frac{1}{2}(v_f + v_i)\Delta t$

- 5 key equations

- Free fall: The motion of a falling object where the only force acting on the object is gravity

Table 1 The Five Key Equations for Uniformly Accelerated Motion

	Equation	Variables found in equation	Variable not in equation
Equation 1	$\Delta\vec{d} = \left(\frac{\vec{v}_f + \vec{v}_i}{2}\right)\Delta t$	$\Delta\vec{d}, \Delta t, \vec{v}_f, \vec{v}_i$	$\vec{a}_{av}$
Equation 2	$\vec{v}_f = \vec{v}_i + \vec{a}_{av}\Delta t$	$\vec{v}_f, \vec{v}_i, \vec{a}_{av}, \Delta t$	$\Delta\vec{d}$
Equation 3	$\Delta\vec{d} = \vec{v}_i\Delta t + \frac{1}{2}\vec{a}_{av}\Delta t^2$	$\Delta\vec{d}, \vec{v}_i, \Delta t, \vec{a}_{av}$	$\vec{v}_f$
Equation 4	$v_f^2 = v_i^2 + 2a_{av}\Delta d$	$v_f, v_i, a_{av}, \Delta d$	Δt
Equation 5	$\Delta\vec{d} = \vec{v}_f\Delta t - \frac{1}{2}\vec{a}_{av}\Delta t^2$	$\Delta\vec{d}, \vec{v}_f, \Delta t, \vec{a}_{av}$	$\vec{v}_i$

If $\Delta d_y = 0$

$\Delta t = \frac{2V_i \sin\theta}{g}$

$\Delta d_x = \frac{v_i^2}{g}\sin 2\theta$

Practice

1. A motorcyclist is travelling at 15.0 m/s [forward] and applies the brakes. The motorcycle slows down at 5.0 m/s² [backward].
   (a) Determine the motorcycle's braking distance. [ans: 23 m [forward]]
   (b) Compare your answer to (a) to the answer for Sample Problem 2. What do these two problems indicate about speeding and traffic safety?
2. A man starts at rest and then runs north with a constant acceleration. He travels 120 m in 15 s. Calculate his acceleration. [ans: 1.1 m/s² [N]]
3. A bus is moving at 22 m/s [E] for 12 s. Then the bus driver slows down at 1.2 m/s² [E] until the bus stops. Determine the total displacement of the bus. [ans: 4.7×10^2 m]
4. In a 100.0 m sprint, a runner starts from rest and accelerates to 9.6 m/s [W] in 4.2 s.
   (a) Calculate the acceleration of the runner. [ans: 2.3 m/s²]
   (b) Calculate the displacement of the runner. [ans: 2.0×10^2 m [forward]]
   (c) The runner runs at a constant velocity for the rest of the race. What is the total time? [ans: 13 s]
5. Two football players separated by 42 m run directly toward each other. Football player 1 starts from rest and accelerates at 2.4 m/s² [right], and football player 2 moves uniformly at 5.4 m/s [left].
   (a) How long does it take for the players to collide? [ans: 4.1 s]
   (b) How far does each player move? [ans: player 1: 20 m; player 2: 22 m]
   (c) How fast is football player 1 moving when the players collide? [ans: 9.6 m/s]
6. A jet lands on a runway at 110 m/s [forward]. When stopping, the jet can accelerate at 6.2 m/s² [backward].
   (a) Calculate the minimum time for the jet to stop. [ans: 18 s]
   (b) What is the minimum safe length for this runway? [ans: 9.8×10^2 m]
   (c) Explain why the runway should be much longer than the minimum safe length.

1) $V_i = 15$ m/s [F]

$\vec{a} = -5$ m/s [F]

a) Δd if $v_f = 0$

$v_f^2 = v_i^2 + 2a\Delta d$

$0 = 15^2 + 2(-5)\Delta d$

$= 225 - 10\Delta d$

$\Delta d = 225$

$= 23$ m

b)?

2) $V_i = 0$

$\Delta t = 15$ s

$\Delta d = 120$ m

$\Delta d = v_i\Delta t + \frac{1}{2}\vec{a}\Delta t^2$

$120 = 0(15) + \frac{1}{2}\vec{a}(15)^2$

$120 = \frac{a}{2}(225)$

$\frac{120}{225} = \frac{a}{2}$

$\cancel{2}\left(\frac{120}{225}\right) = 1.06$ m/s²

$\vec{a} = 1.06$ m/s²

3 a) $\vec{V}_i = 22$ m/s [E]

$\Delta t = 12$ s

$\vec{a} = -1.2$ m/s² [E]

$\Delta d = v_i\Delta t + \frac{1}{2}\vec{a}(\Delta t)^2$

$= 22(12) + \frac{1}{2}(-1.2)(12)^2$

$\Delta d = 264 + (\frac{1}{2})(-1.2)(144)$

$\Delta d = 177.6$ m

4) $V_i = 0$

$V_f = 9.6$ m/s [W]

$\Delta t = 4.2$ seconds

$\vec{a} = ?$

$\vec{\Delta d} = ?$

$V_f = \vec{v}_i + \vec{a}\Delta t$

$9.6 = 0 + a(4.2)$

$\frac{9.6}{4.2} = a$

$\vec{a} = 2.28$ m/s² [W]

b) $\vec{\Delta d} = v_i\Delta t + \frac{1}{2}\vec{a}\Delta t^2$

$\vec{\Delta d} = \frac{2.28}{2}(4.2)^2$

$\Delta d = 20.11$ m [W]

c) $100 - 20.11 = 79.89$ m

$V = 9.6$ m/s

$t = \frac{d}{v}$

$= \frac{79.89}{9.6} =$

$t = 8.3$

$\Delta t = 8.3 + 4.2 = 12.5$ s

5)

1.2 Review

Summary

- The five key equations for uniformly accelerated motion, listed in Table 1 on page 18, use the following variables: displacement, initial velocity, final velocity, acceleration, and time interval.
- Free fall is the motion of an object when it is moving only under the influence of gravity.
- The acceleration due to gravity on Earth is 9.8 m/s² [down].

Questions

1. Upon leaving the starting gate, a racehorse accelerates at a constant 4.1 m/s² [forward] for 5.2 s. Determine the horse's
   (a) displacement
   (b) final velocity KU A
2. An electron travelling at 7.72×10^6 m/s [E] enters a force field that reduces its velocity to 2.46×10^6 m/s [E]. The acceleration is constant. The displacement during the acceleration is 0.478 m [E]. KU A
   (a) Determine the electron's acceleration.
   (b) Determine the time interval over which the acceleration occurs.
3. A police officer at rest at the side of the highway notices a speeder moving at 62 km/h along a straight level road near an elementary school. When the speeder passes, the officer accelerates at 3.0 m/s² in pursuit. The speeder does not notice until the police officer catches up. KU A
   (a) How long will it take for the officer to catch the speeder?
   (b) How far will they move from the position where the officer was at rest?
   (c) Calculate the speed of the police car when the officer catches the speeder. Is this reasonable?
   (d) Now assume that the police officer accelerates until the police car is moving 10.0 km/h faster than the speeder and then moves at a constant velocity until the police officer catches up. How long will it take to catch the speeder? Is this scenario more reasonable than in (c)? Explain your answer.
4. A cliff rises straight up from the Mediterranean Sea. Explain how you could calculate the height of the cliff above the sea using a stopwatch and a small stone. What assumption must you make? MC TI C A
5. A baseball player throws a ball straight up and then catches it 2.4 s later at the same height from which he threw it. Determine
   (a) the initial velocity
   (b) the maximum height of the ball KU A
6. You throw a ball straight up into the air at 18 m/s from a height of 32 m above the ground. KU
   (a) Calculate the time the ball takes to hit the ground.
   (b) Calculate the velocity of the ball when it hits the ground.
   (c) Calculate the maximum height of the ball.
   (d) Explain why you cannot use half the time for (a) to answer (c).
7. A toy rocket starts from rest on the ground and then accelerates at 39.2 m/s² [up] for 5.0 s. KU
   (a) Calculate the velocity of the rocket when the engines stop firing after 5.0 s.
   (b) Calculate the maximum height of the rocket.
   (c) How long will it take the rocket to hit the ground from rest?
8. When a driver is forced to make a panic stop by pressing down on the brake as hard as possible, the car will undergo a large acceleration to stop. MC TI A
   (a) Copy **Table 2** into your notebook. Calculate the braking distances.
   (b) Compare the reaction times, and discuss the implications to the driver and to speeding of using cellphones while driving.

Table 2

	Acceleration (m/s²)	Reaction time (s)	Speed (km/h)	Braking distance (m)
(i)	9.5	0.80	60.0	
(ii)	9.5	0.80	120.0	
(iii)	9.5	2.0	60.0	

Questions

1. Upon leaving the starting gate, a racehorse accelerates at a constant 4.1 m/s² [forward] for 5.2 s. Determine the horse's
   (a) displacement
   (b) final velocity KU A

1a) $\vec{v}_i = 0$

$\vec{a} = 4.1\ m/s^2$

$\Delta t = 5.2\ s$

$\vec{\Delta d} = ?$

$\vec{\Delta d} = v_i \Delta t + \frac{1}{2} a \Delta t^2$

$= 0 + \frac{1}{2}(4.1)(5.2)^2$

$\vec{\Delta d} = 55.432\ m$ [Forward]

b) $\vec{v}_f = v_i + \vec{a}\Delta t$

$= 0 + 4.1(5.2)$

$\vec{v}_f = 21.32\ m/s$ [Forward]

2. An electron travelling at 7.72×10^6 m/s [E] enters a force field that reduces its velocity to 2.46×10^6 m/s [E]. The acceleration is constant. The displacement during the acceleration is 0.478 m [E]. KU A
   (a) Determine the electron's acceleration.
   (b) Determine the time interval over which the acceleration occurs.

$\vec{v}_i = 7,720,000\ m/s$ [E]

$\vec{v}_f = 2,460,000\ m/s$ [E]

$\vec{\Delta d} = 0.478\ m$ [E]

$v_f^2 = v_i^2 + 2\vec{a}_{av}\Delta d$

$2,460,000^2 = 7,720,000^2 + 2a(0.478)$

$2,460,000^2 = 7,720,000^2 + a(0.956)$

$-5.3468 \times 10^{13} = a(0.956)$

$\vec{a} = -5.6011 \times 10^{13}$ [E]

b) $v_f = v_i + \vec{a}\Delta t$

$2,460,000 = 7,720,000 + (-5.6011 \times 10^{13})\Delta t$

$\Delta t = 9.391 \times 10^{-8}\ s$

$2,460,000 = 7,720,000 + (-5.691\times10^{13})\,\Delta t$

$\Delta t = 9.391\times10^{-8}\,s$

Important (Study)

3)

3. A police officer at rest at the side of the highway notices a speeder moving at 62 km/h along a straight level road near an elementary school. When the speeder passes, the officer accelerates at 3.0 m/s² in pursuit. The speeder does not notice until the police officer catches up. T/I A

(a) How long will it take for the officer to catch the speeder?

(b) How far will they move from the position where the officer was at rest?

(c) Calculate the speed of the police car when the officer catches the speeder. Is this reasonable?

(d) Now assume that the police officer accelerates until the police car is moving 10.0 km/h faster than the speeder and then moves at a constant velocity until the police officer catches up. How long will it take to catch the speeder? Is this scenario more reasonable than in (c)? Explain your answer.

3a) Speeder

$v_1 = 62$ km/h

$v_1 = 17.2$ m/s

Cop

$\vec{v}_1 = 0$ m/s

$\vec{a} = 3.0$ m/s²

$\Delta d_{sp} = \Delta d_{cop}$

$\Delta d = \vec{v}_1 \Delta t$ | $\Delta d = v\Delta t + \frac{1}{2}a\Delta t^2$

$\Delta d = 17.2\Delta t$ | $\Delta d = 0 + \frac{1}{2}(3)\Delta t^2$

$17.2\Delta t = 1.5\Delta t^2$

$17.2 = 1.5\Delta t$

$\frac{17.2}{1.5} = \Delta t$

$\Delta t = 11.46$ s

b) $\Delta d = v_1\Delta t$

$= 17.2(11.46)$

$= 198$ m

c) Cop

$v_1 = 0$

$\vec{a} = 3.0$ m/s

$\Delta t = 11.46$ s

$\vec{v}_f = v_1 + \vec{a}\Delta t$

$\vec{v}_f = 0 + 3(11.46)$

$\vec{v}_f = 34.38$ m/s

Not realistic

d) Now: $v_f = 72$ km/h

$v_f = 20$ m/s

time to reach 20 m/s

$v_f = v_1 + a\Delta t$

$20 = 0 + 3\Delta t$

$\Delta t = 6.67$ s

distance covered by cop

$\Delta d = v_i\Delta t + \frac{1}{2}\vec{a}\Delta t^2$

$= 0.3(3)(6.67)^2$

$\Delta d = 66.7$ m

distance covered by speeder

$v = \frac{\Delta d}{\Delta t}$

$\Delta d_s = 17.2(6.67)$

$= 114.8$ m

Gap =

$\Delta d_g = 114.8 - 66.7 = 48.1$ m

$v_{rel} = 20 - 17.2 = 2.8$ m/s

time to close gap

$= \frac{\Delta d}{v_{rel}} = \frac{48.1}{2.78} = 17.3$ s

$\boxed{17.3 + 6.67 = 24 \text{ second}}$

Holy shit

4)

↳ assume no air friction
- assume g is 9.8 m/s² down
- Assume the rock fall straight down
- Assume the surface of the water is flat and uniform

5. A baseball player throws a ball straight up and then catches it 2.4 s later at the same height from which he threw it. Determine

(a) the initial velocity

(b) the maximum height of the ball T/I A

5) Stage 1

$v_1 = ?$

$v_f = 0$

$\vec{a} = -9.8$ m/s² [up]

$\Delta t = \frac{2.4}{2} = 1.2$ s

$v_f = v_1 + \vec{a}\Delta t$

$0 = v_1 + (-9.8)(1.2)$

$v_1 = 11.76\ m/s\ [up]$

b) $\Delta d = v_1 \Delta t + \frac{1}{2} \vec{a} \Delta t^2$

$\Delta d = 11.8(1.2) + (-4.9)(1.2)^2$

$\Delta d = 7.056\ m\ [up]$

6) 6. You throw a ball straight up into the air at 18 m/s from a height of 32 m above the ground. TA

(a) Calculate the time the ball takes to hit the ground.

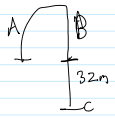
(b) Calculate the velocity of the ball when it hits the ground.

(c) Calculate the maximum height of the ball.

(d) Explain why you cannot use half the time for (a) to answer (c).

a) $v_1 = 18\ m/s\ [up]$

$\Delta d = ?\ + 32m$



A) $v_1 = 18\ m/s\ [up]$

$v_f = 0$

$\vec{a} = -9.8\ m/s^2$

$v_f = v_1 + a\Delta t$

$0 = 18 + (-9.8)\Delta t$

$\frac{-18}{-9.8} = \Delta t$

$\Delta t_1 = 1.84\ s$

B) $v_1 = 0$

$v_f = ?$

$\vec{a} = -9.8\ m/s^2$

$\Delta t_2 = 1.84\ s$

$v_f = 0 + (-9.8\ m/s)(1.84)$

$v_f = -18.032\ m/s\ [up]$

C) $v_1 = -18.032\ m$

$\Delta d = -32m$

$\vec{a} = -9.8\ m/s^2$

$\Delta d = v_1 \Delta t + \frac{1}{2} \vec{a} \Delta t^2$

$-32 = -18.032(\Delta t) + (-4.9)(\Delta t)^2$

$0 = -4.9\Delta t^2 - 18.03\Delta t + 32$

$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$\frac{18.03 \pm \sqrt{(-18.03)^2 - 4(-4.9)(32)}}{2(-4.9)}$

$\frac{18.03 - 30.86}{-9.8} = 1.309$

$\Delta t_3 = 1.309$

$\Delta t = 1.84 + 1.84 + 1.309$

$\Delta t = 5s$

b) $v_1 = -18.032$

$\Delta t = 1.309$

$\vec{a} = -9.8\ m/s^2$

$v_f = v_1 + \vec{a}\Delta t$

$v_f = -18.032 - 9.8(1.309)$

$v_f = -30.86\ m/s\ [up]$

c) $v_1 = 18\ m/s$

$\vec{a} = -9.8$

$\Delta t = 1.84$

$\Delta d = v_1 \Delta t + \frac{1}{2} \vec{a} \Delta t^2$

$= 18(1.84) + (-4.9)(1.84)^2$

$= 16.53m\ [up] + 32m\ [up]$

$\vec{\Delta d} = 48.5\ m\ [up]$

7. A toy rocket starts from rest on the ground and then accelerates at 39.2 m/s² [up] for 5.0 s. TA

(a) Calculate the velocity of the rocket when the engines stop firing after 5.0 s.

(b) Calculate the maximum height of the rocket.

(c) How long will it take the rocket to hit the ground from rest?

a) $v_1 = 0\ m/s$

$\vec{a} = 39.2\ m/s^2\ [up]$

$\Delta t = 5\ sec$

$\vec{v_f} = \vec{v_1} + \vec{a}\Delta t$

$v_f = 0 + 39.2(5)$

$\vec{v_f} = 196\ m/s\ [up]$

b) $\Delta d_1 =$

$v_f^2 = v_1^2 + 2\vec{a}\Delta d$

$v_f = 196\ m/s$ [up]

b) $\Delta d_1 =$

$v_f^2 = v_1^2 + 2\vec{a}\,\Delta d$

$196^2 = 0 + 2\,(39.2)\,\Delta d$

$\Delta d = 490\ m$

$\Delta d_2 = \vec{v}_1 = 196\ m/s$

$\vec{a} = -9.8\ m/s^2$

$\Delta d = ?$

$v_f = 0$

$0^2 = 196^2 + 2\,(-9.8)\,d$

$-38416 = -19.6\,d$

$\Delta d_2 = 1960\ m$

$\Delta d = 2450\ m$

c)